

JAGADEESHWARAN M

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PROFILE

Systems-oriented AI/ML Engineering student experienced in designing and implementing machine learning pipelines across edge CV, geospatial modeling, and LLM-agent architectures. Proficient in Python, modular API construction, and optimizing models for resource-constrained environments. Proven ability to solve practical engineering challenges, demonstrated through active participation in competitive hackathons and open-source software development.

PROJECTS

FISHVISION : Built an offline-first mobile app to identify ~30 Indian fish species using YOLOv8 + MobileNetV3. Developed a React Native frontend with fast local search and filtering based on habitat, region, and family. Designed a scalable Node.js backend with JWT authentication, SQLite storage and image-processing pipeline for future expansion to 2,500+ species. Runs completely offline.

Tech Stack : YOLOv8, MobileNetV3, Python ,OpenCV, React Native, Expo, Node.js, Express.js, JWT, SQLite.

OPENGATE: Built an interactive GATE study portal and agentic RAG tutor featuring a glassmorphic UI and local SQLite progress tracking. Developed a LangGraph routing agent that dynamically coordinates queries between segmented vector databases, a SymPy mathematical engine for exact symbolic calculations, and DuckDuckGo web search. Integrated experiment tracking using MLflow to log prompts, retrieved contexts, and routing steps to DagsHub.

Tech Stack: LangGraph, LangChain, ChromaDB, FastAPI, Python, SymPy, MLflow, DagsHub, DVC, SQLite, JS, CSS.

NEER : AI-Powered Lake Health Monitoring System. Designed an end-to-end geospatial AI system to assess water health of Coimbatore lakes using satellite data (NDWI, NDCI, FAI, MCI, turbidity, thermal). Built time-series regression models to predict BOD, DO, and COD using multi-year Sentinel-2 data. Classified pollution causes and suggested improvement strategies. Visualized results on interactive maps with dynamic filters

Tech Stack : Google Earth Engine, Python, Scikit-learn, Remote Sensing Indices, GeoJSON, React, Flask, leaflet.

ROTUS : Built a real-time object tracking system achieving around 50 fps using Raspberry Pi 5, Google Coral TPU, and ZED 2i camera. Implemented a computer vision pipeline for hardware-accelerated detection with SORT-based multi-object tracking, and stereo disparity-based depth estimation for 3D localization for robotics and edge-AI applications.

Tech Stack : TensorFlow Lite, SSD MobileNet V2, OpenCV, Python, SORT, Linux (Raspberry Pi OS).

EDUCATION

Bachelor of Technology in Artificial Intelligence and Machine Learning
Sri Shakthi Institute of Engineering and Technology, Coimbatore, Tamil Nadu
ABC Matriculation Higher Secondary School, Coimbatore, Tamilnadu

CGPA: 7.41
2023 - 2027
2022 - 2023 (82%)

SKILLS

- **Languages:** Python, SQL, C, HTML, CSS, JavaScript
- **Tools/Frameworks:** Flask, FastAPI, Streamlit, Chainlit, Git, Linux, MySQL, Docker
- **AI/ML:** Scikit-learn, PyTorch, LangChain, LangGraph, deepagents, Langsmith, litellm, Pandas, NumPy, Matplotlib, OpenCV, TensorFlow(Keras), TFLite, Ultralytics(YOLO), MobileNet, Ollama, MLflow, DagsHub

CERTIFICATIONS

- Prompt Engineering **by NVIDIA**
- Data Analytics in Python **by NPTEL**
- Machine learning specialization **by Stanford**
- Deep learning specialization **by Deeplearning.AI**